BenchMatch / Round 1 proposal

Open Innovation

Official static validation passed - assigned NAS execution pending

1. What does the Skill do?
BenchMatch compares documented conditions of two prototype runs and preserves differences, missing values and source conflicts in a reviewed report.

2. Target user and problem
Makers and small workshops keep run notes in scattered files and often repeat a failed test without knowing what changed.

3. Proposed solution
Reconcile run-bound parameters and reported outcomes, expose gaps and prepare a consistent recording checklist for the next test.

4. Core user flow
Identify comparable runs and variables; authorize sources; inspect differences and unknowns; add or clarify a note; approve the report.

5. Substantive difference
The workflow preserves variable-level uncertainty and run identity instead of turning a plausible explanation into an unsupported cause.

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6. UGREEN AI NAS capabilities

File discovery establishes run sources; document extraction and reconciliation recover values and units; optional supported transcription adds a note; approved text saving preserves the comparison.

7. Technical approach

The parent delegates ug-filesystem inventory/verification to file-agent, ug-document qa/reconcile to document-agent, and ug-knowledge transcription/save-text to knowledge-agent. Inputs specify runs, context, variables, units and tolerances. Verify routing before private reads; prefer local processing. Separate approval precedes writes. No equipment control or external API is included. Codex assisted development.

8. Current development status

Implemented: original Markdown workflow, references, synthetic fixtures, concept demo, proposal and local checks. UGOS Skill Creation Tool 0.7.3 and catalog 2.0.0 were used to reconcile authoring; official static reports accompany the release. NAS installation, execution, extraction accuracy and model routing remain untested. Entrant review is pending.

9. Final validation plan

Core one: identify temperature and lot changes while keeping missing humidity unknown. Core two: incorporate a run-linked humidity note and retain conflicting temperatures. Edge: incompatible units or unresolved run identity must prevent an unjustified comparison.

10. Known limitations

Incomplete records and extraction errors limit findings. Comparable differences do not establish causality. Unit mappings require explicit support, and the workflow does not prescribe machine settings or certify an experiment.

Third-party information (excluded from substantive response limit)

UGREEN tools/experts: UGREEN terms; no kit redistributed. No third-party runtime code, dataset or model bundled. Separate demos: Rubik, SIL OFL 1.1. PDF creation: original Python standard-library code. Local checks: pypdf, BSD-3-Clause; PyYAML, MIT. Sources and full disclosures accompany the Skill.